



A Course End Project Report on
Online Quiz Application

A9602 - Object Oriented Programming through Java Laboratory

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Abstract

The Online Quiz Application is a Java-based desktop application developed using Java Swing to provide an interactive platform for conducting multiple-choice quizzes. The application allows users to register and log in using a username and password, select a subject from Java, Python, and C++, and attempt a set of multiple-choice questions.

The application provides a graphical user interface with separate screens for login, registration, subject selection, quiz participation, and result display. During a quiz, users can select one of four available options for each question, while the application automatically evaluates the selected answer and maintains the user's score. After all questions are answered, the final score is displayed to the user.

The application demonstrates the use of Java Swing components, event handling, object-oriented programming concepts, CardLayout-based screen navigation, and array-based quiz data management. It provides a simple and user-friendly approach to conducting subject-based quizzes while demonstrating the practical implementation of GUI programming concepts in Java.

Keywords: Java Swing, Online Quiz Application, Graphical User Interface, Multiple-Choice Quiz, Event Handling, CardLayout, Object-Oriented Programming.

1 Introduction

The Online Quiz Application is a Java-based desktop application designed to provide a simple and interactive platform for conducting subject-based multiple-choice quizzes. The application is developed using Java Swing and provides a graphical user interface through which users can register, log in, select a subject, answer quiz questions, and view their final scores.

The application includes quizzes for three programming languages: Java, Python, and C++. Each subject contains a set of multiple-choice questions with four options. The application evaluates the user's selected answers and calculates the score automatically. This provides an easy way for users to test and improve their knowledge of programming concepts.

The project demonstrates several important Java programming concepts, including GUI development using Swing, event handling, object-oriented programming, arrays, conditional statements, and screen navigation using CardLayout. Different screens are provided for login, registration, subject selection, quiz participation, and result display, making the application organized and easy to use.

The main purpose of developing this application is to demonstrate the practical implementation of Java GUI programming concepts through a functional quiz system. It combines user interaction with automated answer evaluation and score calculation, providing a basic foundation that can be further expanded with features such as databases, larger question banks, user profiles, and performance tracking.

2 Literature Review

Online quiz applications have become a common approach for providing interactive assessment and knowledge evaluation. Traditional quizzes generally depend on paper-based questions and manual evaluation, which can require additional time and effort. Computer-based quiz systems provide an alternative by allowing questions to be presented digitally and answers to be evaluated automatically.

Java provides several features that are suitable for developing desktop-based quiz applications. In particular, the Java Swing framework provides GUI components such as buttons, labels, text fields, password fields, and panels that can be used to create interactive applications. Event handling allows the application to respond to user actions such as button clicks and option selection.

The development of a quiz application also provides an opportunity to apply fundamental programming concepts such as object-oriented programming, arrays, conditional statements, methods, and event-driven programming. In the developed application, these concepts are combined to manage user registration, login, subject selection, question navigation, answer evaluation, and score calculation.

This project focuses on implementing these concepts in a simple desktop-based quiz system. The application provides separate quizzes for Java, Python, and C++, demonstrating how a graphical interface can be combined with program logic to create an interactive learning and assessment tool.

3 Objectives

The main objectives of the Online Quiz Application are:

1. To develop a **user-friendly graphical quiz application** using Java Swing.
2. To implement a basic **user registration and login system** for accessing the application.
3. To provide subject-based quizzes for **Java, Python, and C++**.
4. To display multiple-choice questions with four answer options for each question.
5. To automatically **evaluate the selected answers** and calculate the user's score.
6. To provide a dedicated **result screen** that displays the user's final quiz score.
7. To demonstrate the practical implementation of **Java Swing, event handling, arrays, methods, conditional statements, and CardLayout**.
8. To develop a simple foundation that can be extended in the future with features such as larger question banks, database integration, user profiles, and performance tracking.

“latex

4 Methodology

The Online Quiz Application was developed as a Java-based desktop application using the Java Swing framework. The methodology involves system design, graphical user interface development, user authentication, subject-based quiz management, answer evaluation, and result generation.

The application is organized into functional modules that work together to provide a simple and interactive quiz experience. System architecture, use case analysis, activity flow, and class design are used to describe the organization and behavior of the application.

4.1 System Architecture

The system architecture describes the major components of the Online Quiz Application and the interaction between them. The application consists of a presentation layer, application logic, quiz engine, data structures, and result processing components.

The presentation layer is developed using Java Swing and provides the login, registration, subject selection, quiz, and result screens. The application logic handles screen navigation and user interaction. The quiz engine manages question presentation, answer evaluation, and score calculation.

User credentials and quiz questions are stored in memory in the current implementation. The result component displays the final score after the user completes all questions.

Figure 1 illustrates the overall architecture of the application.

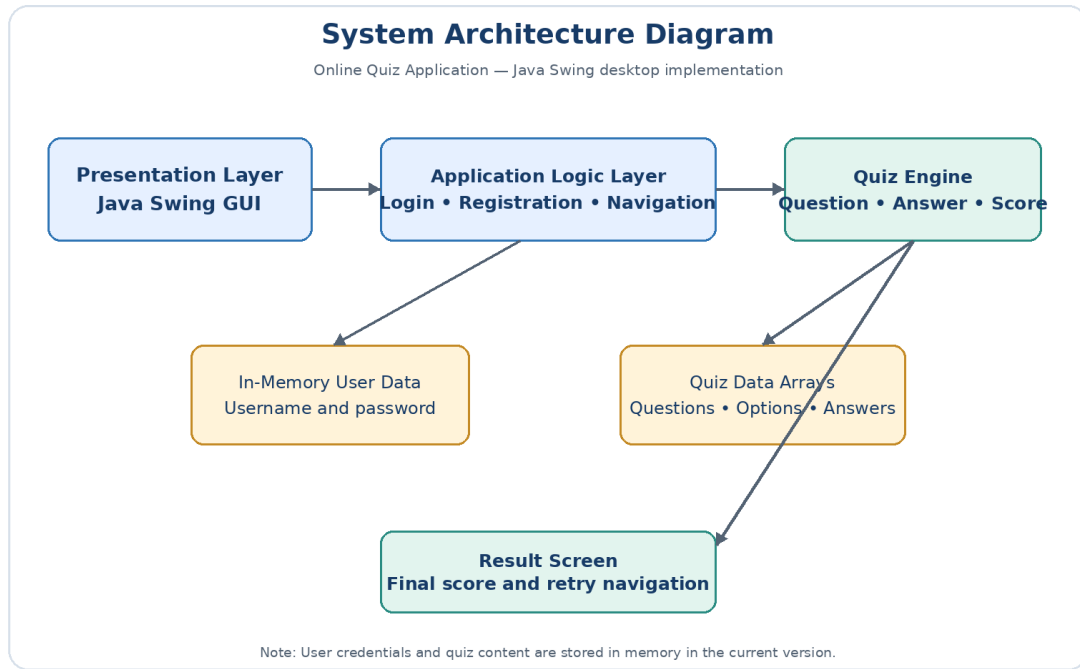


Figure 1: System Architecture of the Online Quiz Application

4.2 Use Case Design

The use case diagram represents the interactions between the user and the Online Quiz Application. The user can register an account, log in, select a programming subject, answer multiple-choice questions, and view the final quiz score.

These interactions define the main functional requirements of the application. The use case design also helps identify the major features that must be implemented through the graphical interface.

Figure 2 presents the primary use cases of the application.

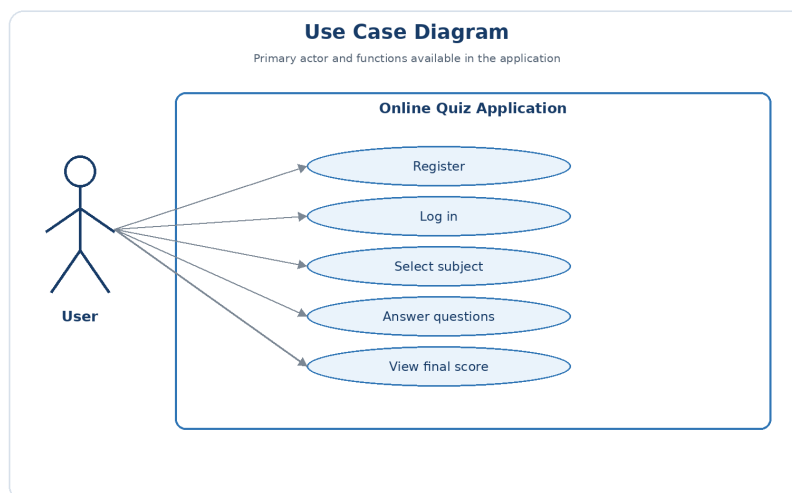


Figure 2: Use Case Diagram of the Online Quiz Application

4.3 System Overview

The application consists of five main screens: Login, Registration, Menu, Quiz, and Result. The user first logs into the application using registered credentials. New users can register by providing a username and password.

After successful login, the user is taken to the main menu, where Java, Python, or C++ can be selected. The corresponding set of questions and answers is then loaded into the quiz screen.

The user answers each question by selecting one of the four available options. The application compares the selected answer with the correct answer stored in the program and updates the score accordingly. Once all questions have been answered, the final score is displayed on the result screen.

4.4 Activity Flow

The activity flow describes the sequence of operations performed during a typical quiz attempt. The application starts by displaying the login screen and validating the entered credentials.

If the credentials are invalid, an error message is displayed and the user can try again. After successful authentication, the user selects a subject and proceeds to the quiz.

The application presents the questions one at a time, evaluates the selected answers, and calculates the final score. The result is displayed after the last question, allowing the user to return to the main menu.

Figure 3 illustrates this sequence of operations.

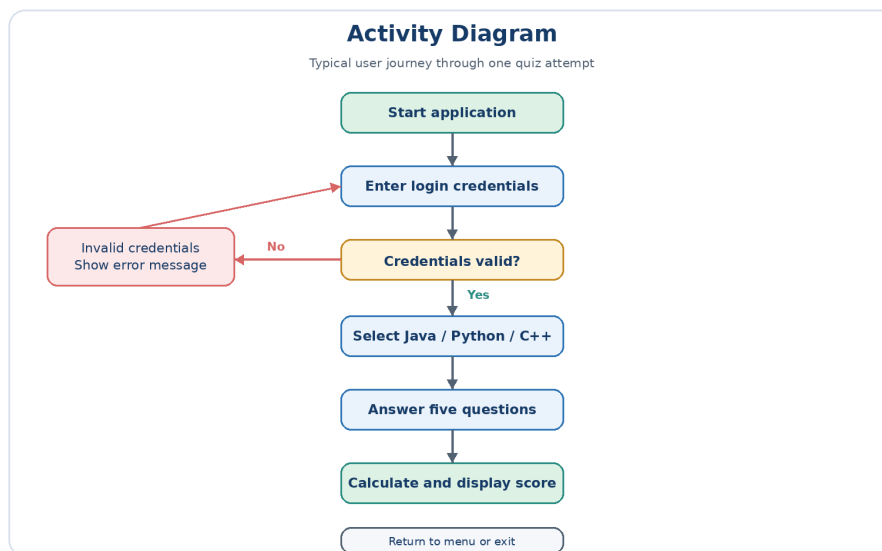


Figure 3: Activity Diagram of the Online Quiz Application

4.5 Main Components

The application is divided into several functional components. Each component performs a specific task and contributes to the overall operation of the quiz system.

4.5.1 Login and Registration Module

The login module provides basic authentication using a username and password. The entered credentials are compared with the registered credentials stored in memory.

If the credentials match, the application displays the main menu. Otherwise, an error message indicating invalid credentials is displayed.

The registration module allows the user to enter a new username and password. Empty username or password fields are rejected. Valid registration information updates the credentials stored by the application.

4.5.2 Subject Selection Module

After successful login, the user is presented with the main menu. The menu provides three quiz options: Java Quiz, Python Quiz, and C++ Quiz. Each option loads the questions and answers associated with the selected subject.

The menu also provides a logout option that returns the user to the login screen.

4.5.3 Quiz Module

The quiz module displays questions and collects the user's answers. Each quiz contains five multiple-choice questions, with four options for every question.

The application maintains the current question index and score using variables. When an option is selected, the application compares its index with the stored correct answer.

If the selected option is correct, the score is incremented. The application then moves to the next question. After the final question is answered, the result screen is displayed.

4.5.4 Result Module

The result module displays a completion message and the user's final score. The score represents the number of correctly answered questions.

The result screen also provides an option to return to the main menu and select another subject.

4.6 Quiz Data Management

The application contains three subject-based question sets: Java, Python, and C++. Each subject contains five questions, four options for each question, and an array containing the correct answer for every question.

When a subject is selected, its corresponding questions, options, and answer array are passed to the quiz module. The question index and score are reset at the beginning of each quiz.

Whenever the selected answer matches the stored correct answer, the score is increased by one. The final score is displayed after all questions have been answered.

4.7 Class Design

The class design describes the responsibilities and relationships of the main software components. The application window manages the graphical interface, screen navigation, question progression, and result display.

The quiz data component represents the question sets, available options, and correct answers. The user account component represents the registration and authentication operations. A custom button component can be used to provide rounded buttons and hover effects.

Figure 4 presents a conceptual class diagram for the application. The class names, attributes, methods, and relationships should be verified against the actual Java source code before the final report is submitted.

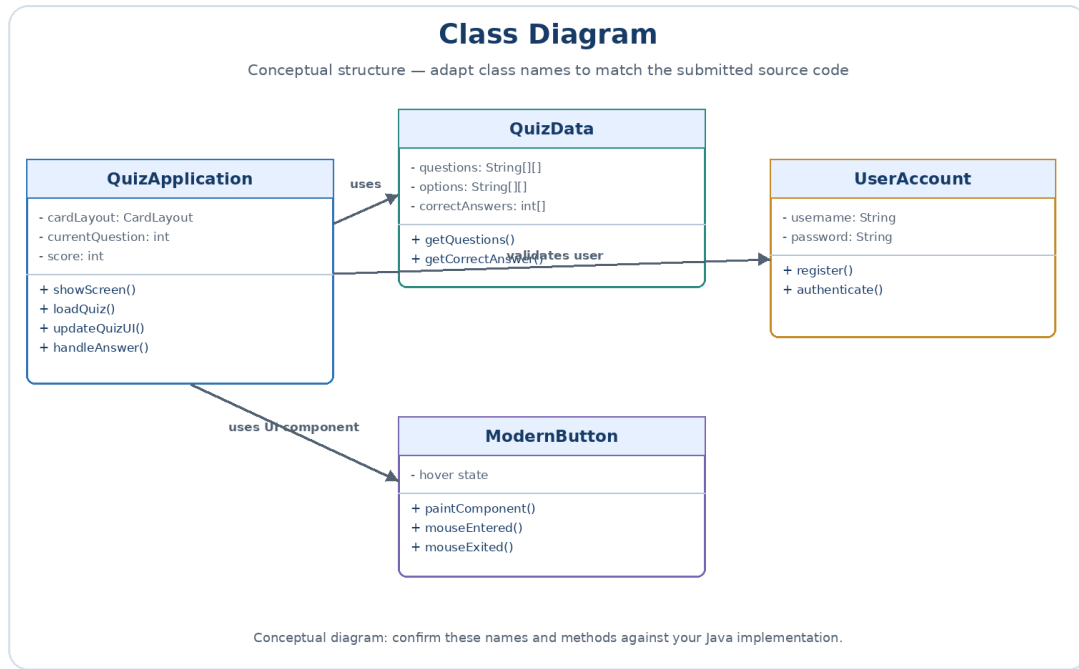


Figure 4: Conceptual Class Diagram of the Online Quiz Application

4.8 User Interface Design

The graphical user interface is developed using Java Swing components, including labels, buttons, text fields, password fields, panels, and dialog boxes.

A consistent visual design is maintained using predefined background colors, text colors, cards, and button hover effects. The application window has a fixed size and is made non-resizable to maintain the intended layout.

A custom `ModernButton` class extends `JButton` to support rounded buttons and hover effects through custom painting and mouse event handling.

4.9 Software Implementation

The application is implemented using Java and the Java Swing framework. The main class extends `JFrame`, which provides the application window. Different screens are implemented as panels and managed using `CardLayout`.

The `loadQuiz()` method loads the selected subject's questions, options, and answers while resetting the question index and score. The `updateQuizUI()` method updates the question and answer buttons displayed on the screen.

The `handleAnswer()` method processes the user's selected answer. It checks whether the selected option matches the correct answer and increases the score when appropriate. The method then advances to the next question or displays the results when the quiz is complete.

The application is launched using `SwingUtilities.invokeLater(code::new)`, ensuring that the graphical interface is created on the appropriate Swing event-dispatching thread.

4.10 Testing Approach

The application can be evaluated by testing the main functional requirements. The tests should cover successful and unsuccessful login attempts, registration validation, subject selection, answer evaluation, score calculation, question progression, and navigation between screens.

Each test should verify that the application produces the expected behavior for the corresponding user action. Any identified issues should be corrected and the relevant tests repeated before the final version is submitted. “

5 Results and Discussion

The developed Online Quiz Application successfully implements the main functions required for conducting subject-based multiple-choice quizzes. The application provides separate screens for user login, registration, subject selection, quiz participation, and result display. The use of Java Swing components provides an interactive graphical interface through which users can easily navigate between these different stages.

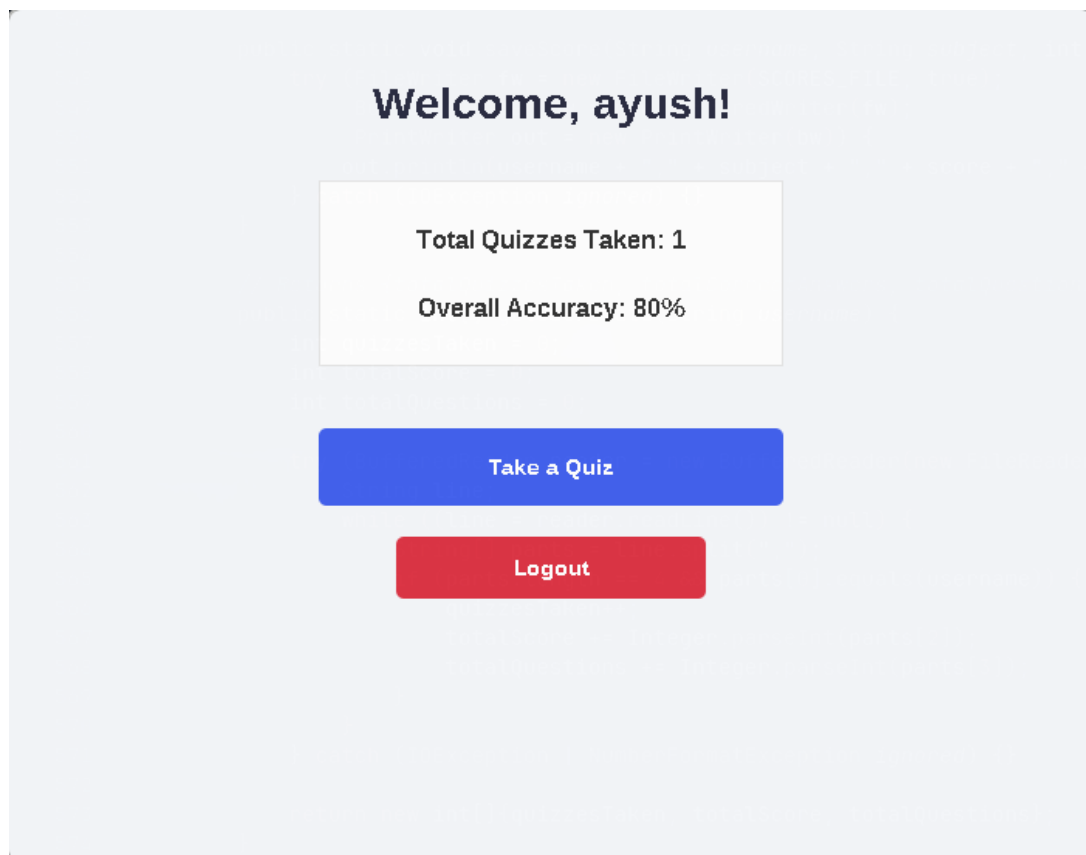
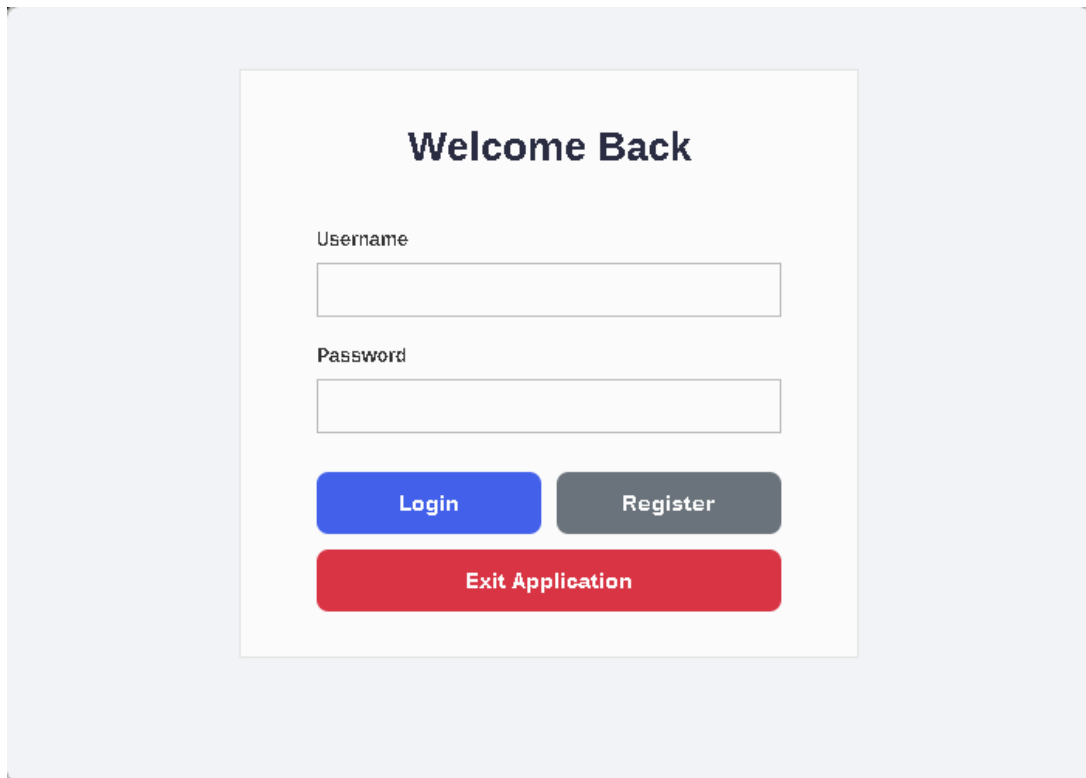
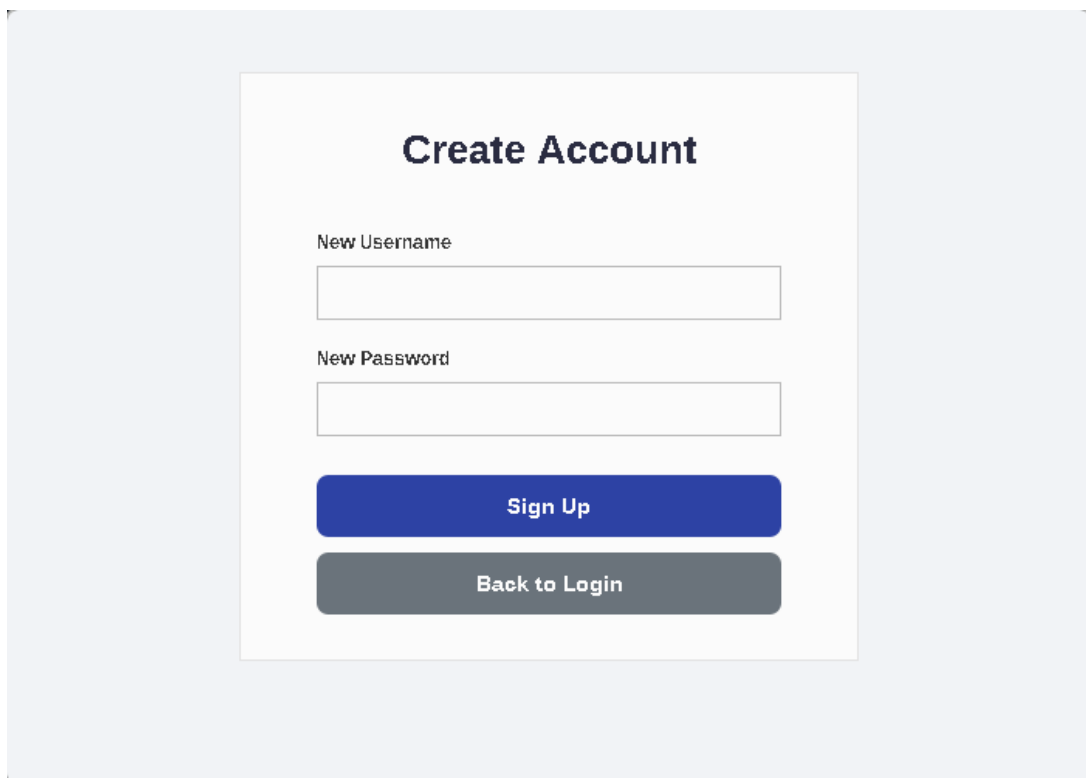


Figure 5: Overall System Flow of the Online Quiz Application



The login screen features a central white card on a light gray background. At the top of the card is the heading "Welcome Back" in bold black text. Below this are two input fields: "Username" and "Password", each with a light gray border. Under the password field are three buttons: a blue "Login" button, a gray "Register" button, and a red "Exit Application" button.

Figure 6: Login Screen of the Online Quiz Application



The registration screen features a central white card on a light gray background. At the top of the card is the heading "Create Account" in bold black text. Below this are two input fields: "New Username" and "New Password", each with a light gray border. Under the password field are two buttons: a blue "Sign Up" button and a gray "Back to Login" button.

Figure 7: Registration Screen of the Online Quiz Application

The login and registration modules provide basic user authentication. Users can register by entering a username and password and subsequently use those credentials to access the

application.

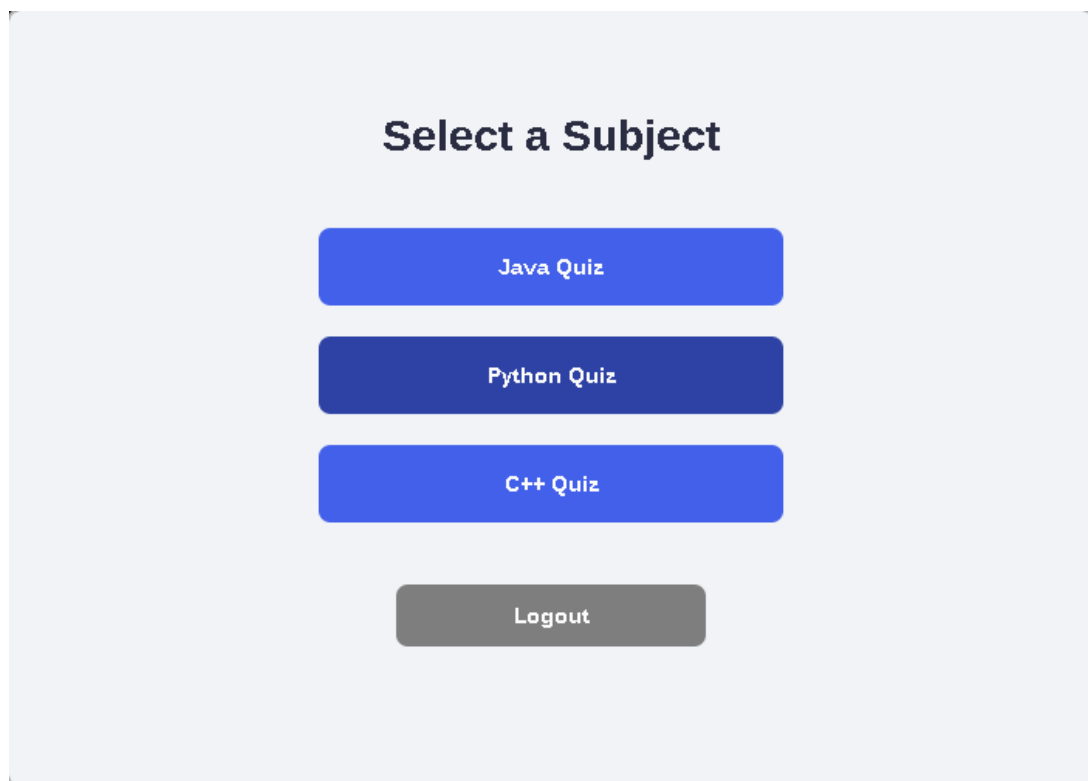


Figure 8: Subject Selection Menu

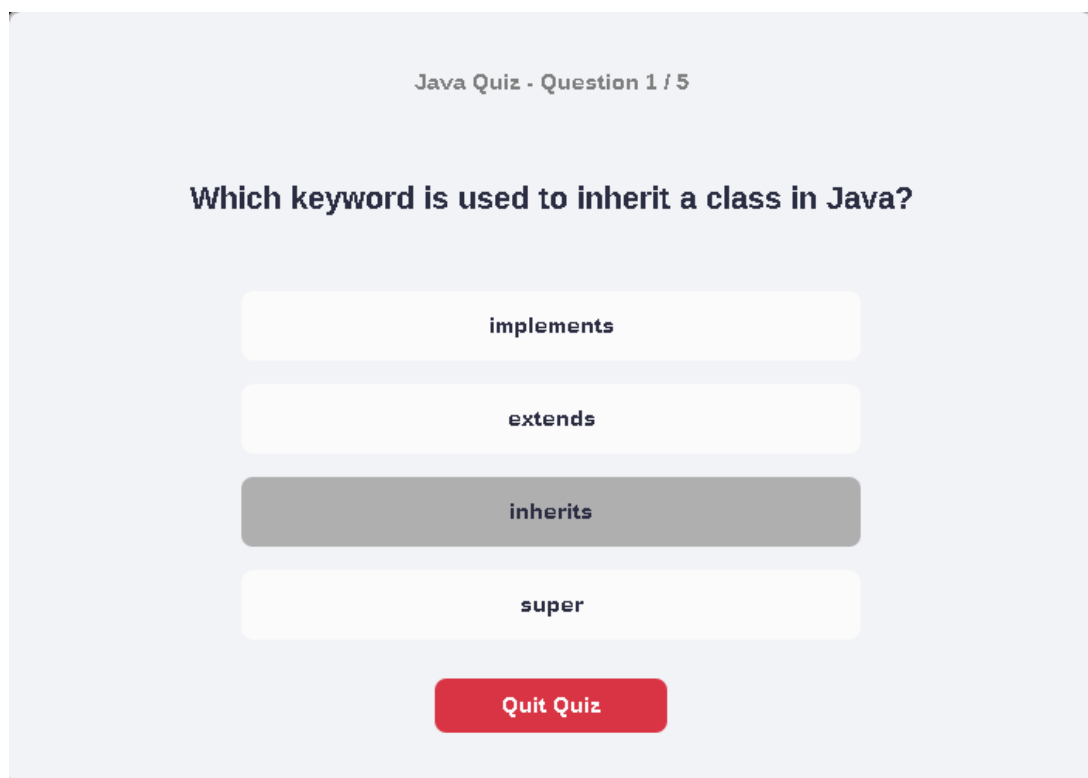


Figure 9: Quiz Screen Showing a Multiple-Choice Question

The subject selection menu allows users to choose between Java, Python, and C++ quizzes. The quiz module displays questions along with four answer options and maintains the current question number and score during the quiz. When a user selects an answer, it is compared with the corresponding correct answer stored in the application. Correct answers increase the score, while incorrect answers do not increase it.

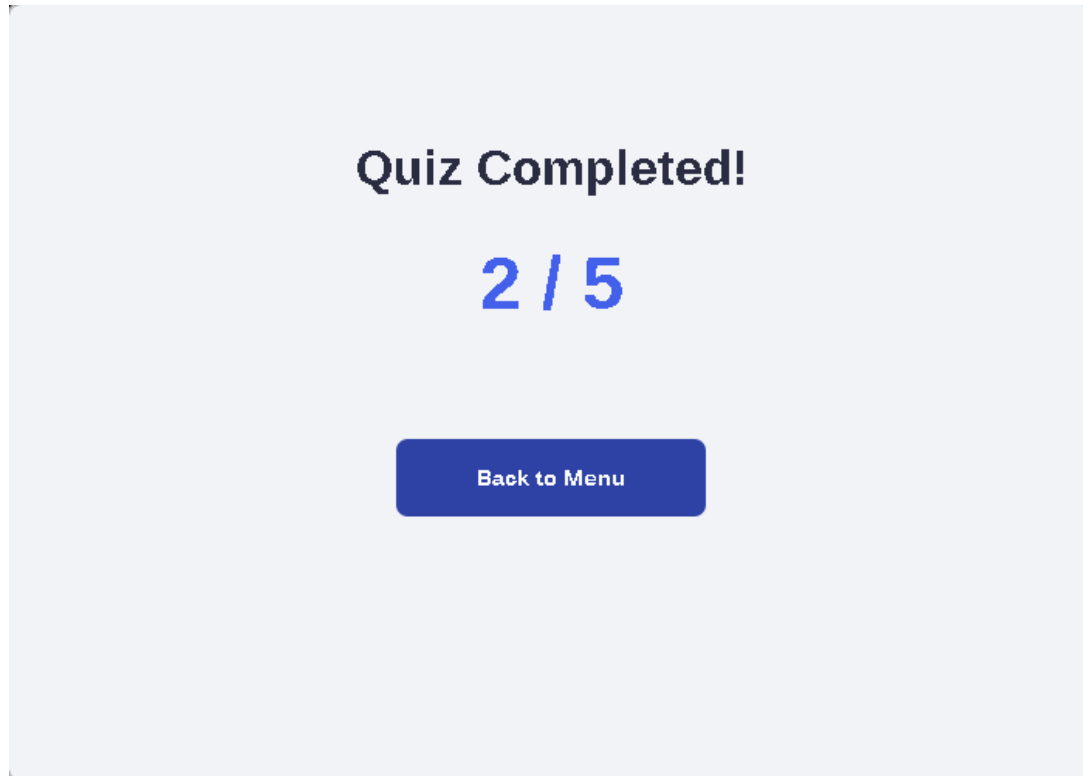


Figure 10: Result Screen Showing the Final Quiz Score

After all five questions are completed, the application displays the final score on the result screen. The graphical interface uses consistent colours, buttons, labels, panels, and hover effects to provide a simple and organized user experience. The use of `CardLayout` allows the different screens to be managed within a single application window without opening multiple windows.

Overall, the application demonstrates the successful integration of Java Swing, event handling, arrays, conditional logic, methods, and object-oriented programming concepts into a functional desktop application. However, the application stores user credentials and quiz data in memory, meaning that the information is not permanently stored after the application is closed. This provides an opportunity for future improvements such as database integration, larger question banks, multiple user accounts, and detailed performance tracking.

6 Mapping POs and SDGs

6.1 Program Outcomes

The Online Quiz Application demonstrates the application of programming, problem-solving, and software development concepts. The following table maps the project with relevant Program Outcomes (POs).

PO	Program Outcome	Contribution of the Project
PO1	Engineering Knowledge	The project applies fundamental programming and software development concepts using Java and Java Swing.
PO2	Problem Analysis	The project involves analyzing the requirements of a quiz system and implementing suitable logic for authentication, question handling, answer evaluation, and score calculation.
PO3	Design/Development of Solutions	A graphical quiz application was designed and developed with separate modules for login, registration, subject selection, quiz participation, and result display.
PO5	Modern Tool Usage	Java Swing, event handling, CardLayout, and graphical interface components are used to develop the application.
PO10	Communication	The application provides clear graphical elements, labels, buttons, and messages to communicate information and instructions to the user.

6.2 Sustainable Development Goals

The project contributes to the following Sustainable Development Goals (SDGs):

SDG	Goal	Contribution of the Project
SDG 4	Quality Education	The application provides an interactive platform for testing programming knowledge through subject-based quizzes.
SDG 9	Industry, Innovation and Infrastructure	The project demonstrates the development of a software-based digital solution using Java and graphical user interface technologies.
SDG 10	Reduced Inequalities	A simple digital quiz platform can provide users with an accessible method of practicing and evaluating their programming knowledge.

7 Conclusion

The Online Quiz Application was successfully developed as a Java-based desktop application using the Java Swing framework. The application provides a simple and interactive platform for conducting multiple-choice quizzes in Java, Python, and C++. It implements important features such as user registration, login authentication, subject selection, question navigation, automatic answer evaluation, score calculation, and result display.

The project demonstrates the practical application of important Java programming concepts, including object-oriented programming, graphical user interface development, event handling, arrays, methods, conditional statements, and CardLayout-based screen navigation. The modular structure of the application makes the different stages of the quiz process easy to understand and use.

Although the current application provides the required basic functionality, it can be further improved by introducing database integration for permanent storage of users and quiz data, larger and dynamically managed question banks, multiple user accounts, user performance tracking, timers, and additional subjects. These enhancements can make the application more scalable and suitable for a larger number of users.

Overall, the project successfully demonstrates how Java programming and GUI development concepts can be combined to create a functional and user-friendly online quiz application.

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